

Software Engineering :

Syllabus



Outline :

- Objectives & Outcomes
- Course Content
- Why it is important
- Teaching Staff
- Evaluation & Assignments
- Resources
- Ethics & Policies.





Objectives & Outcomes

- The aim of this module is to provide the basic concepts for creating a functional and usable software product via complying with the principles of Software Engineering.
- Develop students' critical skills to distinguish sound development practices from ad-hoc practices, judge which technique would be most appropriate for solving large-scale software problems, and articulate the benefits of applying sound practices.



Objectives & Outcomes

- At the end, students are expected to learn and acquire :
 - Software Product is not the outcome of mainly Programming.
 - Different phases for the lifecycle of software development.
 - Work within a team to develop a large software product.
 - Collaboration
 - Communication
 - Management
 - Resolving Conflicts
 - Best Practices and methodologies for software engineering

Content & Plan

Course Content

1. Introduction

- a. Software Development challenges
- b. Software scope
- c. Software Engineering Discipline
- d. Software Methodologies and Related Processes
- e. The Human Side of Software Development
- f. Case studies

2. Software processes

- a. Software process models
- b. Process Activities
- c. Traditional Life Cycle Models
 - i. Waterfall
 - ii. V
 - iii. Phased
 - iv. Evolutionary
 - v. Spiral
- d. Alternative Techniques
 - i. Unified Process
 - ii. RAD
 - iii. JAD
 - iv. Prototyping
- e. Agile Software Engineering Process Models
 - i. Extreme Programming

- ii. Agile Software Development
 - iii. DevOps
 - iv. Site Reliability Engineering (SRE)
 - f. IEEE Standards for Software Engineering Processes and Specifications (optional)
 - g. ISO 12207: Life Cycle Standard (optional)
- ### 3. Requirements engineering
- a. Functional and non-functional requirements
 - b. Requirements engineering processes
 - c. Requirements elicitation
 - d. Requirements specification
 - e. Requirements validation
 - f. Requirements change
 - g. Agile Requirements Engineering (optional)
 - h. Requirements Management Tools (e.g., IBM Rational Doors) (Optional/TP)
- ### 4. System modelling
- a. Modelling Languages (Unified Modelling Language UML, OCL....)
 - b. Context models
 - c. Interaction models
 - d. Structural models
 - e. Behavioural models
 - f. Model-driven architecture
- ### 5. Architectural design
- a. Architectural design decisions
 - b. Architectural views
 - c. Architectural patterns
 - d. Application architectures
 - e. Using Open Source, free, freemium, paid, and Enterprise software components (optional)
 - f. Design Tools (e.g., Sparx Enterprise Architect) (Optional/TP)
- ### 6. Design and implementation
- a. Object-oriented design using the UML
 - b. Design patterns
 - c. Language and platform issues
 - d. Refactoring
 - e. Test Driven Development (TDD) (optional)
 - f. Distributed Development and Agile Methods Scalability (optional)
 - g. Development Tools (Optional)
 - 1. IDEs (e.g., Xcode, Eclipse, IntelliJ IDEA, NetBeans, Microsoft Visual Studio, Atom)
 - 2. Source Control Management (e.g., GitHub)
 - 3. Release Orchestration (e.g., OpenMake)
 - 4. Collaboration (e.g., Jira, Trello, Slack)
- ### 7. Software Testing
- a. Unit Testing
 - b. Integration and System Testing
 - c. Static Confirmation
 - d. Dynamic Testing
 - e. Automated Testing
 - f. Test-driven development
- ### 8. Advanced Topics In Software Engineering

Content & Planning



1. Introduction to Software Engineering
2. Software Engineering Processes & Models
3. Agile Methodologies
4. Requirements Engineering
5. Design : Software Modeling : UML - Part 1
6. Design : Software Modeling : UML - Part 2
7. Design : Software Modeling : UML - Part 3



Content & Planning

8. Software Architecture : Technological Stack + Styles
9. Software Architecture : Principles - SOLID + Decoupling & Cohesion
10. Software Architecture : Design Patterns
11. Software Development : Best Practices
12. Software Development : Case Study
13. Software Testing
14. Maintenance & Integration



Why it is important ?

- In this course , you will :
 - Be taught the process for developing and delivering a **professional** software project.
 - Be given the opportunity to work within a simulated professional environment by working inside a team to deliver a software product.
 - Acquire most of the technical skills, technological tools and competences that you can publish on your CV.
 - Combine all notions and theories learnt in other courses (Programming, Management, Problem Solving, Data Structures...) into solving a concrete problem

Why it is important ?

- In this course , you will :

- Be taught the professional software project.
- Be given the opportunity to work in a professional environment by working inside a company.
- Acquire most of the technical and transversal competences that you can publish on your CV.
- Combine all notions and theories learnt in other courses (Programming, Management, Problem Solving, Data Structures...) into solving a concrete problem.

Software Engineering = CV

Teaching Staff

- **Prof. Okba Tibermacine,**

okba.tibermacine@ensia.edu.dz

Office Hours : Sunday 11:50, Administrative Bloc (You must send an email first)

- **Prof. Imed Bouchrika**

imed.bouchrika@ensia.edu.dz

- **Dr. Fouad Dahak**

fouad.dahak@ensia.edu.dz

- **Dr. Khadidja Chettah**

khadidja.chettah@ensia.edu.dz

Learning Process



Lecture Slides + Lecture Demos are sent to students to have a glimpse about the materials

Attending lectures to understand the materials

Quiz Form will be sent to students to try before the Lab session

Tutorial/LAB Sheet will be sent to students to look into the exercises before the session.

Students must do the Pre-LAB / Pre-Tutorial exercises before the sessions

Attending LAB/Tutorial Sessions.

LAB/Tutorial Solutions are sent on Thursdays/Fridays



Evaluation & Assignments

- Exam : 60%
- Continuous Evaluation : 40%
 - Mid-term : **5 points** :
 - Online Quiz : **3 points** :
 - Team Project : **10 points**
 - Lab Homework, Assignments & Staff Appreciation : **2 points**
 - Challenge Exercises : **2 points** (Extra credit)
 - Penalty of Non-Justified Absence : **-5 points**
- Late Submission :
 - 1 day Penalty of 20%
 - 2 days Penalty of 50%
 - 3 days → Zero mark is given.



Quizzes (3 points)

- Quizzes will be available either :
 - After the lecture and closed at the end of the week.
 - During the lecture only.
 - Only for the students who **attended** the lecture(s)
- Missing **three** quizzes, zero mark will be given directly and all other submissions will not be considered.
- There will be only 8 quizzes
- **Being absent in 5 lectures, ZERO mark will be given.**



Quizzes (3 points)

- Online Quizzes should preferably be done before attending the tutorial sessions.
- Quizzes include easy/medium M.C.Questions aimed to enforce your understanding for the concepts discussed during or related the lecture.
- PLEASE : Don't rush to answer the quiz, take your time, **search for the solutions on the web** and submit your answers when you are certain.
- The quiz will be open during the week of the lecture. (To ensure students will not procrastinate doing their revision and homework)



Group Project (10 points)

- Team Creation :
 - Teams of 5 or 6 members will be created.
 - No teams of 2 or three or four students.
 - You will not be choosing who will work with you ?.
 - The staff will create project teams before the TD session by randomly selecting students in the **same TD Group**.
 - You are **not** allowed to change your team to another team once we settle
 - The aim : You are employed with within a team, learn how to live, integrate and adapt to what differences, conflict you will be having.



Group Project (10 points)

- Team Leader :
 - Up to the team to have a democratic process to select a leader or any other process they like to invent .
 - But we need to have a student name as a Team Leader from week 1
 - Deadline : **02 October 2025**
 - **Form to submit : <https://forms.gle/Xjaba2bGGX4KBvF6A>**



Group Project (10 points)

- Project Topic : Deadline : **Thursday 09 October 2025**
 - Three Options:
 - *The team can come up with an idea for a project to implement.*
 - *Provided it is validated by a staff*
 - *We will give you a list of topics to choose from. **No two teams can choose the same topic from the same group.***
https://docs.google.com/spreadsheets/d/1GgA7Liy1MnfYXnJnVRtTIRUlo-XsHp_1oKCj7QNjLQ/
 - *Students can propose Topics for other teams to implement (Send us your topic by Email as soon possible)*



Group Project (10 points)

- Project Topic : Deadline : **Thursday 09 October 2025**
 - The project topics must :
 - *Feasible within the resources you have*
 - *Will be used by real people .*
 - *It solves an existing problem*
 - *Not over ambitious.*
 - No Restriction on the Technology to be used (But no use of Mobile Technology)

Group Project (10 points)

- Steps for Choosing Project Topic :

1. You consult the list of proposed Topics on the shared Excel File
https://docs.google.com/spreadsheets/d/1GgA7Liy1MnfYXnJnVRtTIRUlo-XsHp_1oKCj7QNjLQ/edit?usp=sharing
2. If you like one from the list, provided that no other teams from your group chose this project → The team leader needs to send us an email to assign their team to the topic.
3. If you like to propose another topic, use the document template provided and submit it using the Form provided.
 - i. If the project is validated and approved, it will be assigned to your team and added to the Excel file for other teams to potentially choose.
 - ii. If not validated, you are forced to choose from the list before the deadline.

Template : https://docs.google.com/document/d/1BPPvVPl2YoBY7jB5tBA8c2u8oRAUdQg367GP8KC8_K4/edit

Form to Submit : Sent Later.



Group Project (10 points)

- Marking Per Student or Per Team :
 - Two marks :
 - Team Mark : We will give one Mark per Team
 - Student Mark = Team Mark $\times$ Penalty Coefficient
 - You work all as a team, you either all win or all lose.
 - If other students are not contributing ? it is not fair that he/she gets the same mark as a working person ? :
 - There will be an individual evaluation per members.
 - Students not contributing, not attending team meetings, Lab sessions, lectures will be penalized.



Group Project (10 points)

- Marking Schema :
 - Continuous Evaluation (TD and TP) : **2 points**
 - Weekly Reporting and Briefing
 - Reaching Milestones
 - Making Continuous Deliverables and respecting deadlines.
 - Final Delivery : **8 points**

Group Project (10 points)

- Marking Schema :

- Continuous Evaluation : **2 points**
- Final Delivery : **8 points**
 - *Functional Software : 2.5 points (Non functioning software $\Rightarrow$ 0pts)*
 - *There are real* users of your software : 1 points*
 - *Design and Implementation : 1 points*
 - *Demonstration : 1 points*
 - *Smooth Collaboration : 0.5 points*
 - *Documentation : 0.5 points*
 - *Reporting and Management : 0.5 points*
 - *Non-Functional Requirements : 1 points*

**Real users means not fake, bots, or ENSIA students or relatives*

Exceptional Project

⇒ 20 points

- Exceptional Project :
 - The team developing a professional software having an exceptional impact
 - Considerable* number of **active real users**.
 - Fully functional
 - Professional Branding and Packaging
 - Solves an existing and true problem
 - Students of the selected team, will get the full mark for :
 - All the quizzes (Provided they don't miss three quizzes)
 - The group project
 - Only one team will be selected

**Real users means not fake, bots, or ENSIA students*



Exams & Midterms

- There will be Questions in the Midterm or the Exam being :
 - Taken from the Optional Exercises
 - Asked inside the lecture
- For the marking procedure, to ensure fairness, each exercise/question is marked by the **same lecturer** for all students.
- If you are incorrect for a given answer, zero mark should be given straight away. However, we may subjectively give some points in appreciation to the effort made by the student. But, we are not required nor you have the right to claim points for incorrect or partially correct answers.

Important Dates



- **29 September 2025** : Initial Phase for Creation of Teams
- **02 October 2025** : **Strict Deadline** for Creation of Teams
- **09 October 2025** : Deadline for Group Project Topics
- **06 November 2025** : Submission of Software Requirements Specifications.
- **08-15 November 2025** : Mid-Term Exams Week
- **04 December 2025** : Submission of Sprint 1
- **Week of 14 December 2025** : Presentations of Delivered Version : Sprint 1
- **08 January 2026** : Final Submission of Group Project
- **Week : 11 January 2025** : Project Validation Week



Challenge Questions

(2 points of extra credit)

- Meant for you to learn how to solve difficult problems that you may face in the future and you can not find their solutions on the web freely.
- Solutions for the challenge Exercises will be submitted via Google Classroom when the Assignments are created. **(It will be created without prior notice with only a one day deadline or subjected to quota)**
- Solving the exercises is optional, but you are encouraged to do so.



Lab & Tutorial Sessions

- Exercises of Prefix :
 - **C** : to be solved in class where you will be given some time to try to solve them.
 - **A : Assignments/homework** that you need to solve within a week deadline. You need to submit your solution via Google Classroom
 - **P** : Are optional exercises to be done at home after class. Discuss with your colleagues about the possible solutions. Feel free to bring your solutions to the lecturer to discuss in class when convenient.
- Note that all **P exercises** are **potential exam** questions that they **will** be included into the final exams.



Tutorial Sessions

- Tutorial Sessions (90 Minutes / Week) will serve as :
 - An opportunity for students to discuss in teams concepts related to the lecture
 - Meeting sessions where teams brief their lecturer about their progress.
 - Mostly, exercises are team exercises (**Make sure to return the chairs/tables as they were**)
 - The use of mobile phones is totally prohibited.



Lab Sessions - TP

- Lab Sessions (90 Minutes / Week) will serve as :
 - Practice Materials related to the lecture to get better understanding.
 - An opportunity for students to work and collaborate on their projects.
 - Meeting sessions where teams brief their lecturer about their progress.
 - Lecturers can demonstrate new technologies or platforms to students
- **Important Note :**
 - Lab Session time should never be wasted to install software or configure them.
 - Lab instructors would offer guidance and recommendations on coding strategies, principles, tips and explanations. But don't expect us to write the code or **debug it**. (As it can be time consuming and unfair to other students seeking legitimate assistance)



Absence & Penalties

- LAB Sessions :
 - Students being absent more than allowed, law will be strictly applied and students will be excluded.
 - Students need to coordinate with the **administration** and LAB staff about their absences.
- Lecture Sessions
 - Students are **advised** to attend all lectures.
 - If you miss the lecture, don't expect LAB staff to repeat lecture materials.
 - Students who are were absent for a valid reason, should email only the lecturer (no need for the administration)
 - Table for Logging all absences : [LINK_HERE...](#)



Absence & Penalties

- Being late :
 - Students coming 10 minutes late, will be allowed in class but they will be considered **ABSENT**
- Absence Penalty Points :
 - **-0.25 Point per Non-Justified Absence** (LAB Sessions OR Lectures)
 - Lab session = one session of 90 minutes
 - A max of **5 points** can be deducted from the Continuous Mark.
- More than 5 absences :
 - All of their points in the bonuses are disregarded.
 -



Attending Lectures

- Can the lectures be substituted by Youtube videos given by Top Scientists from MIT and Stanford ?
- Are lectures about telling you what's inside the book that you can read on your own ?
- Deliberate missing of lectures :
 - Would contribute to create a lazy atmosphere among the students
 - Accumulation of pressure and stress before the exam.
- Of course, You **may** learn a lot on your own without attending lectures, but for 7 modules within a semester, you are putting yourself at a risk.



Exceptional Cases

- Exceptional Progress : Mid-term vs. Exam Marks :
 - Students who scored below 10 in the midterm whilst their score above ($>$) 15 in the exam, their midterm mark will be the same as the exam.
- Exceptional Projects :
 - Students who come up with an Exceptional Project with a true impact, they will get the full mark for the continuous mark.
- Exceptional Cases of Justified Absences:
 - Students who have missed considerable number of sessions, please always contact us for assistance when needed to catch up.



Resources

- Course Textbook :
Ian Sommerville. "Software Engineering, Pearson Edition", 2015.
Marco Tulio Valente. "Software Engineering Modern approach", 2024
- Materials from other universities:
 - University of California, San Diego .<https://cseweb.ucsd.edu/classes/su08/cse70/schedule.php>
 - University of Toronto : <http://www.cs.toronto.edu/~matt/csc444/lectures.html>
 - California State University : <https://www.cs.csub.edu/~gordon/3350/>
 - Carnegie Mellon University : <https://www.cs.cmu.edu/~aldrich/courses/413/>



Policies

- The use of **Mobile Phones inside the classroom is prohibited for Chat and non-educational purposes**. Disciplinary measures would be taken as it is considered as a disrespect to the lecturers and to the school.
- Chewing Gum + Coffees are strictly prohibited inside the classrooms
- Penalty for Late Submissions will be applied strictly. Any submission that we cannot open or unzip, zero mark will be given directly. Submission by email email is never acceptable.
- If you need assistance or have any questions, always speak directly to the lecturer, lab staff and make use of the weekly office hours.